AY: 2022-23 Onwards	J.B.INSTITUTE OFENGINEERING AND TECHNOLOGY (UGC Autonomous)	B. Tech-CSE IVYear- I Sem			
Course Code: L41MD	COMPUTER VISION	L	T	P	C
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Pre-Requisites:

1. Linear Algebra, Calculus, Probability and Statistics
2. Data Structures

Course Objectives:

The students should be able to

1. Review image processing techniques for computer vision.
2. Understand multicamera views.
3. Understand Hough Transform and its applications to detect lines, circles, ellipses.
4. Understand three-dimensional image analysis techniques.
5. Understand motion analysis and study some applications of computer vision algorithms.

Module 1:

Introduction to Computer Vision:

Computer Imaging Systems, Digital Image Formation and Low-Level Processing, Overview and State-of-the-art, Fundamentals of Image Formation.

Transformation: Orthogonal, Euclidean, Affine, Projective, etc. Fourier Transform, Convolution and Filtering, Image Enhancement, Restoration, Histogram Processing.

Module 2:

Depth Estimation and Multi-Camera Views:

Perspective, Binocular Stereopsis: Camera and Epipolar Geometry; Homography, Rectification, DLT, RANSAC, 3-D Reconstruction Framework; Auto-Calibration

Module 3:

Feature Extraction:

Edges - Canny, LOG, DOG; Line Detectors (Hough Transform), Corners - Harris and Hessian Affine, Orientation Histogram, SIFT, SURF, HOG, GLOH, Scale-Space Analysis- Image Pyramids and Gaussian Derivative Filters, Gabor Filters and DWT.

Module 4:

Image Segmentation:

Region Growing, Edge Based Approaches to Segmentation, Graph-Cut, Mean-Shift, MRFs, Texture Segmentation; Object Detection

Module 5:

Motion Analysis:

Background Subtraction and Modelling, Optical Flow, KLT, Spatio-Temporal Analysis, Dynamic Stereo; Motion parameter estimation.

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